PES UNIVERSITY
OS Assignment problem 4
PES2UG24CS362

Github repo:
https://github.com/PranavBalebail/PES2UG24CS362-pes-vcs/tree/main

Outputs
Codes in github given above

Analysis Questions
Phase 5
Q5.1: Implementation

- .pes/ changes: Update the HEAD file to store the new branch path (e.g., ref: refs/heads/new-branch).
- Working Directory: Replace current files with those from the target commit's tree.
- Complexity: Ensuring atomicity. You must update the Index and Working Directory simultaneously while preventing the loss of uncommitted local changes.

Q5.2: Detecting "Dirty" Conflicts

Compare hashes between the current HEAD, the Index, and the Target Branch:

- If Index Hash != HEAD Hash, the file is modified.
- If Target Hash != HEAD Hash, the checkout would change that file.
- Conflict: If both are true and Index Hash != Target Hash, the operation must refuse to avoid overwriting uncommitted work.

Q5.3: Detached HEAD

- Committing: Commits work normally, but HEAD stores the raw hash. No branch reference moves forward.

- Recovery: If you switch away, these commits become "orphaned." They can be recovered using the reflog to find the lost hashes and point a new branch at them.

Phase 6

Q6.1: Mark-and-Sweep

- Algorithm: Start at all branch/tag references ("roots"). Recursively "mark" every reachable commit, tree, and blob. "Sweep" (delete) any object in the store not marked.
- Data Structure: A Hash Set to store visited object IDs.
- Visit Estimate: For 100,000 commits, you'd likely visit 500,000+ objects (commits plus their associated trees and blobs).

Q6.2: Race Conditions

- The Risk: A GC process might see a brand-new blob, realize no branch points to it *yet*, and delete it just as a concurrent commit process is trying to link it to a new tree.
- The Fix: Git uses a grace period (usually 2 weeks). GC only deletes unreachable objects that haven't been modified or created recently, ensuring "in-flight" data is safe.

Screenshots:

Phase 1

1A ./test_objects

```
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ make test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B

```
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$
```

Phase 2

2A ./test_tree

```
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

Phase 3

3A pes init → pes add → pes status sequence

```
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ echo "hello" > file1.txt
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ echo "world" > file2.txt
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ ./pes add file1.txt file2.txt
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  README.md
  untracked:  tree.h
  untracked:  .gitignore
  untracked:  index.c
  untracked:  test_tree
  untracked:  object.c
  untracked:  test_sequence.sh
  untracked:  test_objects.c
  untracked:  test_objects
  untracked:  index.h
  untracked:  pes.c
  untracked:  pes.h
  untracked:  Makefile
  untracked:  tree.c
  untracked:  commit.c
  untracked:  commit.h
  untracked:  test_tree.c
```

3B cat .pes/index showing the text-format index

```
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776696292 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776696321 6 file2.txt
```

Phase 4

4A pes log output with three commits

```
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ export PES_AUTHOR="Pranav <PES2UG24CS362>"
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ echo "hello" > file1.txt
./pes add file1.txt
./pes commit -m "first commit"
Committed: d98588c833af... first commit
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ echo "world" >> file1.txt
./pes add file1.txt
./pes commit -m "second commit"
Committed: b9596f76c587... second commit
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ echo "!" >> file1.txt
./pes add file1.txt
./pes commit -m "third commit"
Committed: 632524efb552... third commit
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ ./pes log
commit 632524efb55261337492e5dcad6925413ae3a53d9ad795e0186077bef29339e7
Author: Pranav <PES2UG24CS362>
Date:   1776696446

    third commit

commit b9596f76c587e2468465bd16805f8c9a53e255554bff4abd91a50f19c31a04b9
Author: Pranav <PES2UG24CS362>
Date:   1776696440

    second commit

commit d98588c833af5ac8df7dcf8e55c223d02530eb81dc7ed6a136823bbc241ded82
Author: Pranav <PES2UG24CS362>
Date:   1776696430

    first commita
```

4B find .pes -type f | sort showing object growth

```
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/49/a197433e6697894bb449eb7fa6fea8744f7a7067792e99cc2d92374b289806
.pes/objects/63/2524efb55261337492e5dcad6925413ae3a53d9ad795e0186077bef29339e7
.pes/objects/b9/596f76c587e2468465bd16805f8c9a53e255554bff4abd91a50f19c31a04b9
.pes/objects/d9/8588c833af5ac8df7dcf8e55c223d02530eb81dc7ed6a136823bbc241ded82
.pes/objects/db/2e16c92bec2bd11723ec4fcfab26dbe59b06136c27d204f2891f2795a56e99
.pes/objects/de/62c80dfbd1163d8b4ee3273c30129e26225ec5521afcf599594b0e2959611d
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/f5/902ee73b6aa25436f687067fc4cc0cb481cf416740d3c3c3ed6d9c4de7b6c2
.pes/objects/fe/76325aa5521b207ebe01e12fd8e9e3abf030cacd5398e3744a3a56a81ad1bd
.pes/refs/heads/main
```

4C cat .pes/refs/heads/main and cat .pes/HEAD

```
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
632524efb55261337492e5dcad6925413ae3a53d9ad795e0186077bef29339e7
ref: refs/heads/main
```

Final screenshot)

Make integration

```
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 0f2c01e75179... Initial commit

Log after first commit:
commit 0f2c01e75179d1296771c520d96dbf12f2c23973cbfde9326d638c166638b489
Author: Pranav <PES2UG24CS362>
Date:   1776696524

    Initial commit


--- Second Commit ---
Committed: 11a00432a3d5... Update file.txt

--- Third Commit ---
Committed: 0ed2c2d51af7... Add farewell

--- Full History ---
commit 0ed2c2d51af790f041b113424f07fe32d8d4ca723d287253929edf2a4068e429
Author: Pranav <PES2UG24CS362>
Date:   1776696524

    Add farewell
```

```
    Add farewell

commit 11a00432a3d556cd70a35a38e1f9e83c020ca48afb17ac9e0981da6413876e9d
Author: Pranav <PES2UG24CS362>
Date:   1776696524

    Update file.txt

commit 0f2c01e75179d1296771c520d96dbf12f2c23973cbfde9326d638c166638b489
Author: Pranav <PES2UG24CS362>
Date:   1776696524

    Initial committial commitES2Q


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
0ed2c2d51af790f041b113424f07fe32d8d4ca723d287253929edf2a4068e429

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/0e/d2c2d51af790f041b113424f07fe32d8d4ca723d287253929edf2a4068e429
.pes/objects/0f/2c01e75179d1296771c520d96dbf12f2c23973cbfde9326d638c166638b489
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/11/a00432a3d556cd70a35a38e1f9e83c020ca48afb17ac9e0981da6413876e9d
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
pranav@panchajanya:~/PES2UG24CS362-pes-vcs$
```